ANDRÉ RICARDO SCHUMACHER TREIB

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PROFESSIONAL SUMMARY

Senior full-stack software engineer with **12 years** of experience, operating at **tech-lead / staff level**: sole architect of multiple production systems shipped end-to-end across **Next.js**, **React Router v7 (Remix)**, **TypeScript** and **Node.js**. Deep working knowledge of **system design**, **event-driven architecture**, **distributed-systems reliability**, **clean architecture**, **TDD** and the full testing pyramid. Joined a US-based startup as its **first hired engineer**, shaped product roadmap and engineering culture, and have spent the last 5 years on English-first remote teams — moving fluently between deep IC work, **mentoring engineers** and product-facing decision-making.

Now specialized in **applied AI / LLM engineering**: I design and operate **autonomous AI-agent workflows over event-driven architecture**, a **custom MCP server**, **RAG pipelines** and **idempotent LLM pipelines with LLM-as-judge evals** — shipped safely into a regulated domain with human-in-the-loop guardrails. Most recently, I championed and built — **solo, from zero to production in 21 days** — a **real-time voice AI agent** (streaming ASR, semantic endpointing, speculative RAG, LLM tool-calling over MCP, streaming TTS — **~0.9 s to first audio**). A recognized **AI-assisted-engineering power user**, I pair purpose-built tooling with rigorous **unit, integration and E2E** coverage to compress scopes that used to take **~4 weeks into ~3 days** without sacrificing quality.

TECH SKILLS

- **AI & Data:** autonomous AI-agent workflows over EDA, real-time voice AI pipelines (streaming ASR, semantic endpointing, VAD & barge-in, streaming TTS, Web Audio / AudioWorklet), custom MCP server, Mastra orchestration, LLM integration (OpenAI / Anthropic / Gemini / Orq.ai), RAG pipelines, vector DBs (Qdrant) & embeddings, speculative retrieval, document chunking & OCR parsing, prompt engineering, reasoning-based orchestration, idempotent AI pipelines, cross-vendor LLM-as-judge evals, confidence-threshold auto-execution, Claude Code power user.
- **Languages & Frameworks:** TypeScript, Node.js, React, Next.js, Remix (React Router v7), HTML, CSS; working knowledge of C#, Kotlin, Rust, Go and Python.
- **Infrastructure:** PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, Redis, Docker, serverless architecture (AWS Lambda Functions), event-driven architecture, microservices.
- **Observability & DevOps:** Full CI/CD pipelines (GitHub Actions / CircleCI), structured logging & telemetry (New Relic), Playwright E2E, Kubernetes overlays, cloud deployment (AWS / Azure), secrets & environment management.
- **Testing & Practices:** TDD; unit, integration, E2E and regression testing; Clean Architecture, SOLID, DDD, design patterns, OOP & functional programming, event-driven systems.

EDUCATION

- Post-graduation, AI Engineering Specialization — **UniPDS** (In progress)
- Post-graduation, Software Engineering Specialization — **UNISINOS** (2021)
- Bachelor, Systems Analysis & Development — **FADERGS** (2017 – 2020)
- IT Technical Course — **Alcides Maya Technical School** (2013 – 2014)
- Native Portuguese, fluent English, and intermediate Spanish

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PROFESSIONAL EXPERIENCE

Senior Software Engineer 04/2022 — Present
House Numbers

First engineer hired at a US-based, AI-native mortgage SaaS; now de facto tech lead of its AI platform.

- Championed and built **Zora Voice**, a production **real-time voice AI agent** for loan processors — **solo, zero to production in 21 days**: streaming **ASR**, **semantic endpointing**, two-stage **barge-in**, **speculative RAG** over an event-sourced **Qdrant** index, multi-vendor **LLM tool-calling over MCP** and streaming **TTS** — **~0.9 s to first audio**, plus a **4-lane quality system** with a **cross-vendor LLM-as-judge** battery
- Sole **architect** of flagship systems — an **event-driven Alerts platform** that removed a third-party SaaS dependency, the **AI-agent workflow engine** (48 production workflows, **~115k LLM runs/month**) and **Title Order-Out** automation; **#2 contributor** to the 16-service monorepo: **~3,500 commits (16% of history)**, **322 PRs, 92% merged**.
- Designed the company's core AI: an **LLM-driven loan-processing agent** over **event-driven architecture** with a **custom MCP server** and idempotent pipelines, saving **12+ hours per loan** — shipped safely via **human-in-the-loop** approvals and **confidence-threshold auto-execution**; invented the team-wide **"reason-then-execute"** pattern.
- Team's **primary code reviewer** and reliability owner: **227 peer PRs reviewed**, resolved production **OOMs, memory leaks and cache stampedes**, and sustained **~1:1 test-to-source coverage** with TDD.
- Acted as **de facto tech lead** — led the investor-demo sprint, owned production pilot rollouts, mentored engineers and **authored company-level technical architecture materials for external executive review**.

Senior Software Engineer 10/2021 — 03/2022
Avenue Code

Embedded with a San Francisco client across a Next.js SaaS.

- Designed and built a **Next.js + serverless SaaS** (TypeScript, PostgreSQL, MongoDB, Auth0, AWS Lambda) with **GitHub Actions CI/CD**; defined **requirements and milestones**, onboarded **3** engineers, and delivered my first professional **Rust** engagement.

Mid Software Engineer; Senior Software Engineer 07/2020 — 10/2021
Eldorado Research Institute

Full-stack engineer on a visual-computing and machine-learning product; co-owner of delivery.

- Built a React, Node.js and Kotlin interface integrated with **ML inference modules** using **TDD and Clean Architecture** end-to-end, and served as **Product Owner** for the first time — owning the product and roadmap.

Mid Software Engineer; Account Manager 02/2018 — 07/2020
MachadoTI

Technical reference who helped an IT-infrastructure business break into software delivery.

- Delivered outsourced projects across **C#, PHP, React, Node.js, Ionic, SQL and NoSQL** for **10+ clients**, while owning **team, cost and deadline management** plus client communication.

Junior Software Engineer; Mid Software Engineer; Scrum Master 04/2014 — 04/2020
Digicounter

Implemented the agile methodology into the company.

- Drove the **full refactor and redesign** of the core vehicle-monitoring SaaS and shipped a mobile app serving **200+ vehicles** and **1,000+ passengers/day**, including an **ETA algorithm**.